Status	Finished
Started	Thursday, 15 October 2026, 6:17 PM
Completed	Thursday, 15 October 2026, 6:20 PM
Duration	3 mins 34 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It only works with integer arrays
b. It requires external libraries
c. It is less efficient than custom sorting algorithms
d. It sorts data out of the box efficiently

The correct answer is: It sorts data out of the box efficiently

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Linear Search
b. Binary Search
c. Bubble Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **3**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It uses the heap data structure
b. It bubbles up the largest element in each pass
c. It recursively sorts subproblems
d. It divides the list into sublists

The correct answer is: It bubbles up the largest element in each pass

Question **4**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By using the bubble-up method
b. By breaking the input into smaller parts and merging them
c. By sorting data in a single pass
d. By comparing elements sequentially

The correct answer is: By breaking the input into smaller parts and merging them

Question **5**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,70,80,90,120,150,200,300,400,470
b. 80,150,200,470,40,90,300,400,70,120
c. 40,80,90,150,200,300,400,470,70,120
d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **6**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. All of the above
- b. Duplicates
- c. Searching
- d. Selection

The correct answer is: All of the above

Question **7**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Quick sort
- b. Selection sort
- c. Heap sort
- d. Merge sort

The correct answer is:
Merge sort

Question **8**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It builds a sorted array one element at a time
- b. It repeatedly finds the minimum element and moves it to the sorted part of the list
- c. It divides the list into two halves, sorts each half, and then merges them
- d. It compares adjacent elements and swaps them if necessary

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **9**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort is less efficient for large lists
- b. Bubble Sort does not guarantee sorted order
- c. Bubble Sort is difficult to implement
- d. Bubble Sort cannot handle duplicate elements

The correct answer is: Bubble Sort is less efficient for large lists

Question **10**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Searching
- b. Sorting
- c. Merging
- d. Complexity

The correct answer is:
Complexity

Question **11**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Heap Sort
- b. Merge Sort
- c. Quick Sort
- d. Bubble Sort

The correct answer is: Bubble Sort

Question **12**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes the data unsorted
b. It complicates the selection of items
c. It slows down the process
d. It makes it easier to select items based on their relationship to the rest of the items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **13**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. Each part is sorted independently
b. The parts are merged without sorting
c. Each part is searched for a specific element
d. The parts are ignored

The correct answer is: Each part is sorted independently

Question **14**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Sorting data in a non-sequential manner
b. Bubbling up the largest element to its correct position with each pass
c. Being the most efficient sorting algorithm
d. Using the divide-and-conquer approach

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **15**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting is rarely used in programming
b. Sorting decreases the efficiency of selection operations
c. Sorting makes it harder to search for items
d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly